

RISC-V TPU指令集 V1.0 修改人: 孟成真 20260430																				
		31	30	29	28	27	26	25	24:20 RS2	19:15 RS1	14	13	12	11	10	9	8	7	6:0 OPCode	
Tensor Memory Instruction	TLOAD	Msk	RowMajor	ColMajor	EStride[4]			MemBaseAddr[U48]	RowSel[U16], MemBaseAddr[U48]	Stride[32], TRD[32]	0	0	0	0	0	0	0	0	0001011	
	TGATHERB	Msk	0	0	0			RowSel[U16], MemBaseAddr[U48]	TRSD[32], TRD[32]	TRSD[32], TRD[32]	0	0	0	0	1	0	0	0		
	TGATHERRB	Msk	0	0	BShape[4]			Row[U8], Col[U8], MemBaseAddr[U48]	TRSD[32], TRD[32]	TRSD[32], TRD[32]	0	0	0	0	1	0	0	1		
	TSTORE	Msk	0	0	BShape[4]			RowSel[U16], MemBaseAddr[U48]	TRSD[32], TRD[32]	TRSD[32], TRD[32]	0	0	0	0	1	0	1	0		
	TSTORE	Msk	RowMajor	ColMajor	EStride[4]			MemBaseAddr[U48]	Stride[32], TRS1[32]	Stride[32], TRS1[32]	0	0	0	1	0	0	0	0		
	TSCATTERRB	Msk	0	0	0			RowSel[U16], MemBaseAddr[U48]	TRSD[32], TRS1[32]	TRSD[32], TRS1[32]	0	0	0	1	1	0	0	0		
	TSCATTERRB	Msk	0	0	BShape[4]			Row[U8], Col[U8], MemBaseAddr[U48]	TRSD[32], TRS1[32]	TRSD[32], TRS1[32]	0	0	0	1	1	0	0	1		
	TSCATTERRB	Msk	0	0	BShape[4]			RowSel[U16], MemBaseAddr[U48]	TRSD[32], TRS1[32]	TRSD[32], TRS1[32]	0	0	0	1	1	0	1	0		
Tensor Multiplication Instruction	TGEMM	0	0	EF	SAT	ACC	BTrans	ATrans	0	TRSD[32], TRS1[32]	1	1	1	0	0	0	0	0		
	TACCSET	0	0	0	0	0	0	0	0	TRSD[32]T	1	1	1	0	0	0	0	1		
	TACCGET	0	0	0	SAT	0	RM3[3]	0	TRD[32]T	0	1	1	1	0	0	0	1	0		
Tensor Arithmetic Instruction	TADD	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	0	0	1		
	TSUB	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	0	1	0		
	TMUL	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	0	1	1		
	TDIV	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	1	0	0		
	TMAX	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	1	0	1		
	TMIN	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	1	1	0		
	TAND	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	0	1	1	1		
	TOR	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	1	0	0	0		
	TXOR	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	1	0	0	1		
	TSLL	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	1	0	1	0		
	TSRL	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	1	0	1	1		
	TSRA	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	1	1	0	0		
	TMERGE	0	0	0	0	0	0	0	TRD[32]T	TRSD[32], TRS1[32]T	0	0	1	0	1	1	1	1		
	TNOT	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	0	0	0		
	TABS	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	0	0	1		
	TNEG	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	0	1	0		
	TSORT	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	0	1	1		
	TRSQRT	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	1	0	0		
	TEXP	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	1	0	1		
	TRCP	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	0	1	1	0		
TRELU	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	1	0	0	0			
TRELU6	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	1	0	0	1			
TSIGMOID	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	1	0	1	0			
TTANH	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	1	0	1	1			
TGELU	0	0	0	0	0	0	0	TRD[32]T	TRS1[32]T	0	0	1	1	1	1	0	0			
TCVT	0	OV3[3]			RM3[3]			max[32], TRD[32]T	min[32], TRS1[32]T	0	0	1	1	1	1	1	1			
Tensor Mask Instruction	TMMV	0	0	0	0	0	0	0	TRD[32]M8	TRS1[32]M8	0	1	0	0	0	0	0	0		
	TMNOT	0	0	0	0	0	0	0	TRD[32]M8	TRS1[32]M8	0	1	0	0	1	0	0	0		
	TMAND	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32]B, TRS1[32]M8	0	1	0	0	0	0	0	1		
	TMNAND	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32]B, TRS1[32]M8	0	1	0	0	1	0	0	1		
	TMOR	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32]B, TRS1[32]M8	0	1	0	0	0	0	1	0		
	TMNOR	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32]B, TRS1[32]M8	0	1	0	0	1	0	1	0		
	TMXOR	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32]B, TRS1[32]M8	0	1	0	0	0	0	1	1		
	TMNXOR	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32]B, TRS1[32]M8	0	1	0	0	1	0	1	1		
	TMEO	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32], TRS1[32]T	0	1	0	1	0	0	0	0		
	TMNEQ	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32], TRS1[32]T	0	1	0	1	1	0	0	0		
	TMLT	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32], TRS1[32]T	0	1	0	1	0	0	0	1		
	TMGE	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32], TRS1[32]T	0	1	0	1	1	0	0	1		
	TMLE	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32], TRS1[32]T	0	1	0	1	0	0	1	0		
	TMGT	0	0	0	0	0	0	0	TRD[32]M8	TRSD[32], TRS1[32]T	0	1	0	1	1	0	1	0		
Tensor Reduce Instruction	TREDMAX	Msk	0	0	0	0	Broadcast	RowOnly	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	0	0	0	0	0		
	TREDMIN	Msk	0	0	0	0	Broadcast	RowOnly	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	0	0	0	0	1		
	TREDSUM	Msk	0	0	0	0	Broadcast	RowOnly	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	0	0	0	1	0		
	TREDAVG	Msk	0	0	0	0	Broadcast	RowOnly	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	0	0	0	1	1		
	TPOLMAX	Msk	0	0	0	0	/4	/2	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	1	0	0	0	0		
Tensor Construction Instruction	TPOLMIN	Msk	0	0	0	0	/4	/2	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	1	0	0	0	1		
	TPOLSUM	Msk	0	0	0	0	/4	/2	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	1	0	0	1	0		
	TPOLAVG	Msk	0	0	0	0	/4	/2	RowOffset[32], TRD[32]	ColOffset[32], TRS1[32]	0	1	1	1	0	0	1	1		
	TFILL	0	0	0	0	0	0	0	RowStep[16], ColStep[16], TRD[32]	Value	1	0	0	0	0	0	0	0		
	TFILLLINE	0	0	0	0	0	0	0	Step[16], TRD[32]	Value	1	0	0	0	0	0	1	0		
	TBCAST	0	0	0	0	0	ColOnly	RowOnly	RowIdx[U16], ColIdx[U16], TRD[32]	TRS1[32]	1	0	0	0	0	0	1	0		
Tensor Control Instruction	TSHIFT	0	0	0	0	0	0	ZeroFill	RowOff[16], ColOff[16], TRD[32]	TRS1[32]	1	0	0	0	0	0	1	1		
	TTRANS	0	0	0	0	0	ColRev	RowRev	TRD[32]	TRS1[32]	1	0	0	0	0	1	0	0		
	TSETMASK	0	0	0	Diag	LowerTri	UpperTri	Invert	TRD[32]M8	Up[16], Bottom[16], Left[16], Right[16]	1	0	0	1	0	0	0	0		
	TFENCE	0	0	0	0	0	0	Reg	Timeout $\neq$ 0	TRS1[32]	1	0	1	RD $\neq$ 0						
	TCANCEL	0	0	0	0	0	0	0	0	0	1	0	1	0						
Tensor IPC Instruction	TSEND	0	0	0	0	0	0	0	Hart[U29], Slot[U3]	TRS1[32]	1	1	0	0	0	0	0	0		
TRECV	0	0	0	0	0	0	0	0	TRD[32]	Slot[U3]	1	1	0	0	0	0	0	1		